Extended reading — Week 01: MDPs and Bandits

October 6, 2026

How to use this list

Read in the order given. The first paper establishes the modern statistical framing of the bandit problem; the second is the algorithm you will implement; the third is the modern Bayesian counterpoint; the fourth is the contextual generalisation that powers production matching systems; the fifth is the unifying analysis that makes all of the above feel like one theory.

1 Robbins (1952) — Some aspects of the sequential design of experiments

Bull. Amer. Math. Soc. 58 (1952), 527–535.

Summary. Robbins frames the two-armed bandit as a problem of *minimising expected loss relative to an oracle*. He proves that a simple "play the winner" strategy is asymptotically consistent and outlines the trade-off between exploration and exploitation that has structured the entire field since. The paper predates Lai & Robbins by 33 years and is short enough to read in a sitting.

Why it matters for this week. Robbins is the historical anchor for the regret framework you proved on UCB1. Reading it makes the modern $O(\log T)$ guarantees feel less like notation games and more like a slow accumulation of the right definitions.

> *"We consider the question of how, on the basis of accumulating experience, to choose between two methods of treatment so as to maximize the expected number of 'successes' or, equivalently, to minimize the expected loss...*

> *The problem so formulated is statistical in character because the underlying parameters are unknown."*

The two-line setup is striking: "maximise successes" immediately becomes "minimise loss vs. the unknown best", i.e. regret. The entire 1952 paper is built on this dual.

2 Auer, Cesa-Bianchi & Fischer (2002) — Finite-time analysis of the multi-armed bandit problem

Machine Learning 47, 235–256.

Summary. Introduces UCB1, UCB1-Tuned, and UCB2, and proves *finite-time* regret bounds of the form $O((K \log T)/\Delta)$. Before this paper most bandit guarantees were asymptotic; the finite-time bound made bandits a practical tool. The proof is a careful Hoeffding + union-bound argument that you reproduced in the notes.

Why it matters for this week. This is the paper whose bound you plotted in Figure 3.2 of the notes. The proof technique — decompose regret into expected suboptimal pulls, bound the deviation events, sum the geometric series — recurs across every UCB-style algorithm in this curriculum.

> *"We show that, when the rewards are bounded, simple confidence-bound based policies achieve, uniformly over time, logarithmic regret. ... The constants are explicit and small enough that the algorithms can be considered for practical purposes."*

"Explicit and small enough" is the contribution. The asymptotic theory of Lai–Robbins was 17 years old; turning it into deployable code required the finite-time bound to be tight in the gap Δ_a.

3 Agrawal & Goyal (2012) — Analysis of Thompson sampling for the multi-armed bandit problem

COLT 2012, arXiv:1111.1797.

Summary. First proof that Thompson sampling achieves $O((K \log T)/\Delta)$ frequentist regret on Bernoulli bandits, matching UCB1 up to constants. The technique — bounding the probability that a sub-optimal arm's posterior sample exceeds the optimal sample — is the template for every subsequent Thompson analysis. Russo & Van Roy (2014, NeurIPS) later sharpened the constants via the information ratio.

Why it matters for this week. Thompson sampling is the algorithm you should reach for in production matching, and now you have a regret bound that justifies the recommendation. The paper also empirically demonstrates TS's superiority on the Yahoo! click-log dataset — the same data Li et al. (2010) used for LinUCB.

> *"The conceptual simplicity of Thompson Sampling, along with its strong empirical performance, makes it an attractive alternative… However, despite being proposed in 1933, theoretical understanding of this algorithm has remained elusive."*

"Theoretical understanding remained elusive" for 79 years. The lesson: empirical performance can survive without proofs for a long time, but proofs are what enable derivative work (variance-reduction tricks, off-policy correction, fairness constraints).

4 Li, Chu, Langford & Schapire (2010) — A contextual-bandit approach to personalized news article recommendation

WWW 2010.

Summary. Introduces LinUCB and the offline replay evaluation technique (now a precursor to modern OPE). Deployed at Yahoo! Front Page; the replayer compares any new policy against logged behaviour without an A/B test. The paper is the genealogical root of every production contextual-bandit system today.

Why it matters for this week. The synthetic insurance / ATS matchers you will build in later weeks are direct descendants of this paper's architecture. Read it for the engineering more than the maths: offline replay, hybrid linear models, regularisation, online updating with Sherman–Morrison.

> *"Using only six weeks of randomly displayed events, our `LinUCB` algorithm doubled the click rate over the editorial control. This rate remained stable for 18 months in continuous A/B test… "*

The bandit improvement was *stable for 18 months*. That is the single most important sentence in applied bandits: a properly-explored linear policy does not need re-training as the world drifts because the per-step posterior update is already an online learner.

5 Russo & Van Roy (2014) — Learning to optimize via information directed sampling

NeurIPS 2014, arXiv:1403.5556.

Summary. Introduces the *information ratio* $\Psi_t = (\mathbb{E}_t[\Delta_t])^2 / I_t(A^\star; (A_t, R_t))$ — expected instantaneous regret squared, divided by mutual information about the optimal action gained by the next pull. Bayesian regret is bounded by $\sqrt{\bar{\Psi} \cdot H(A^\star) \cdot T}$ where $H(A^\star)$ is the prior entropy of the optimum. The argument unifies UCB and TS proofs and motivates *information-directed sampling* (IDS) as an explicit information-gain maximiser.

Why it matters for this week. If you only learn one piece of theory from Week 1, learn the information ratio. It makes precise the intuition that "a good exploration strategy explores in proportion to information gain". It also explains why TS works: TS implicitly bounds its information ratio, so its Bayesian regret is automatically near-optimal.

> *"We propose a new approach to balancing exploration with exploitation: at each period, sample an action so that the squared expected regret is small relative to the information acquired about the optimal action."*

The information-theoretic re-framing turns "how much to explore?" into a *quantity you can compute*. Subsequent work (Kirschner & Krause 2018; Hao, Lattimore & Szepesvári 2022) builds linear and Gaussian-process information-ratio bounds.

Reading exercises

After working through the five papers above, attempt the following.

1. **Robbins, sequel.** Read Lai & Robbins (1985, *Adv. Appl. Math.*) and reproduce the change-of-measure step that delivers the 1/KL lower bound. Where in the proof does the assumption of *consistency* of the algorithm enter? Could the bound be tightened for a restricted family of algorithms (say, only UCB-type)?

2. **Auer et al., gap dependence.** Derive the regret bound for UCB1 when the gaps satisfy $\Delta_a = \Delta$ for all sub-optimal a. The bound becomes $O((K-1)\log T/\Delta)$; verify this against your empirical regret curve in the workbook. What is the implicit worst case as $\Delta \to 0$? (Hint: minimax-style $O(\sqrt{KT})$ regret of MOSS.)

3. **Agrawal–Goyal, beyond Bernoulli.** Extend Thompson sampling to Gaussian rewards with a Normal-Gamma prior. Write out the conjugate update and code it. Compare its regret with UCB1 on a 5-arm Gaussian bandit with unit-variance rewards and means in $\{0, 0.1, \ldots, 0.4\}$.

4. **Li et al., hybrid LinUCB.** Read §3.2 of the paper. Hybrid LinUCB shares a global coefficient β across arms and a per-arm θ_a. Write down the joint Bayesian linear-regression update for $(\beta, \{\theta_a\})$. Why does hybrid LinUCB dominate disjoint LinUCB when arms have few observations each?

5. **Russo–Van Roy, information ratio for two-arm Bernoulli.** Compute the information ratio Ψ of Thompson sampling on a two-arm Bernoulli bandit with prior Beta$(1, 1)$ at $t = 1$. Verify the bound $\Psi \le K/2 = 1$ that Russo–Van Roy prove for general K-armed bandits.

Connections to later weeks

The exact same machinery powers the following later weeks:

- *Week 11 (contextual bandits + OPE)* extends LinUCB to neural bandits and integrates the off-policy estimators from `research/bandits_matching_sota.md` §3.
- *Week 12 (RL for ranking)* treats the slate-bandit problem as a contextual bandit over actions of size k, with position-bias correction.
- *Week 17 (exploration / meta)* brings the information ratio back in the form of information-directed sampling and Bayesian deep exploration.
- *Week 22 (CMDPs)* adds safety constraints to the OFU principle you learned this week, producing the Constrained Policy Optimisation / Lagrangian recipe.

The thread of "optimism in the face of uncertainty" (OFU) runs through the curriculum from Week 1 to Week 24. Memorising the bandit version makes the deep version readable.

Sourced from

`research/bandits_matching_sota.md` sections 1–3. For the broader historical context, see Lattimore & Szepesvári, *Bandit Algorithms* (Cambridge UP, 2020), Chapters 7–19, and Sutton & Barto, Chapter 2.